AI Practice Portfolio

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Three small, working prototypes built while preparing for this role — each tackles a real, current problem created or exposed by how AI is actually being adopted in organizations right now. Full code, real captured outputs, and complete writeups: github.com/midhesh/ai-practice-portfolio

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1. Decision Log — an MCP server that proactively catches contradictions

Problem: organizational decisions go stale silently, and this has worsened as AI chat tools became part of daily work — a growing share of real reasoning now happens inside disposable, one-off AI conversations that leave no institutional trace once the tab closes. The outcome persists; the *why* doesn't.

What I built: an MCP (Model Context Protocol) server where **log_decision** doesn't just store what it's told — before saving, it proactively searches existing decisions for semantic overlap using real TF-IDF + cosine-similarity vector search (built from scratch, zero dependencies) and flags a possible contradiction unprompted. Decisions can explicitly supersede earlier ones, keeping history auditable.

Real, verified result: a near-duplicate decision was correctly flagged at 34% similarity before saving. A later decision explicitly superseded the original — a query for the original topic correctly surfaced it marked **SUPERSEDED**, pointing to the current decision instead of resurfacing dead reasoning as live.

2. PromptGrade — a self-patching eval harness

Problem: AI tool adoption is accelerating faster than any team's ability to verify it's reliable. Prompts get trusted because they “seemed fine” on a few manual tries; failures on edge cases surface only after rollout.

What I built: a harness that grades AI responses against defined rubrics across realistic test cases, and closes the loop agentically with **synthesize_patch()**, which derives a targeted prompt fix directly from a failing case's own rubric — no human-authored patch required. The loop is genuinely observe → diagnose → patch → re-verify.

Real, verified result: on a 5-case triage suite, prompt v1 scored 4/5, missing a real failure mode (a recurring complaint misread as one-off); one added instruction brought it to 5/5. To prove the loop generalizes, I ran it live against a held-out case never seen before: a trivial complaint with “URGENT!!!” in the subject line. The current-best prompt failed it; the auto-synthesized patch passed it on the first re-try.

3. AI Pilot Scorecard — with a responsible-use risk gate

Problem: MIT's 2025 “State of AI in Business” study found roughly 95% of generative AI pilots fail to reach production or deliver measurable ROI — the report's own conclusion is that most pilots never had a defined success bar, so results get judged by feeling after the fact. (source)

What I built: an interactive tool that forces success criteria to be defined before a pilot runs, then grades the real result against that bar. It goes further with a **responsible-use risk gate**: if a pilot touches irreversible actions, PII, or client-facing exposure with no human review, the verdict is capped at *Iterate* regardless of how good the metric looks, with a specific suggested next step.

Real, verified result: an 80%-actual-vs-70%-target pilot that would otherwise score *Scale* was correctly capped to *Iterate* the moment “irreversible action” and “no human review” were both flagged — catching a good number hiding a bad decision, which a metrics-only scorecard would miss.

Why these three, together

Each addresses a different failure mode in how AI actually gets adopted inside organizations: **losing the reasoning behind decisions and letting stale ones linger** (Decision Log), **trusting AI workflows without verifying or fixing them** (PromptGrade), and **scaling a pilot on a good number while missing the risk underneath it** (Pilot Scorecard). None of these are about AI being more powerful — they're about AI adoption being more disciplined, which is the actual work this role is describing.